

Derivation of Least Squares Estimators for Linear Regression

Given a data set of n points $x_i, y_i, i \in \{1, 2, \dots, n\}$, we seek to find the line that fits best. Suppose that the line is given by the equation:

$$y = mx + b$$

Notice that y_i is the y coordinate value of the i -th point, and suppose, $\hat{y}_i = mx_i + b$ is the y coordinate value predicted by the linear regression line for x_i . As always, to fit the line, we want to minimize the difference between these quantities, namely, $e_i := y_i - \hat{y}_i$ for every i . Two points here: Firstly, minimizing their sum $\sum_{i=1}^n e_i$, is the same as minimizing all of the e_i terms for any i . Secondly, these terms or their sum are not differentiable. Hence, to make them differentiable, we have to square them. So, the quantity we want to minimize is

$$\mathbf{E} = \sum_{i=1}^n e_i^2 = (y_i - \hat{y})^2$$

Therefore, we seek to determine the values of m and b that minimize the **Sum of Squared Errors**, we will denote as follows:

$$\mathbf{E} = \sum_{i=1}^n e_i^2 = (y_i - \hat{y})^2 = \sum_{i=1}^n (y_i - (mx_i + b))^2.$$

Please note that here m and b are the unknowns and x_i and y_i are the known values given in the n points in the data set.

Step 1: Compute Partial Derivatives

[Following steps as per what we studied in high school calculus]

To minimize $\mathbf{E}$, we take the partial derivatives with respect to m and b , and set them to zero.

Partial Derivative with respect to m :

$$\begin{aligned} \frac{\partial \mathbf{E}}{\partial m} &= \sum_{i=1}^n 2(y_i - (mx_i + b))(-x_i) \\ &= -2 \sum_{i=1}^n x_i(y_i - mx_i - b). \end{aligned}$$

Setting this to zero:

$$\sum_{i=1}^n x_i y_i - m \sum_{i=1}^n x_i^2 - b \sum_{i=1}^n x_i = 0.$$

Partial Derivative with respect to b :

$$\begin{aligned}\frac{\partial \mathbf{E}}{\partial b} &= \sum_{i=1}^n 2(y_i - (mx_i + b))(-1) \\ &= -2 \sum_{i=1}^n (y_i - mx_i - b).\end{aligned}$$

Setting this to zero:

$$\sum_{i=1}^n y_i - m \sum_{i=1}^n x_i - nb = 0.$$

Step 2: Solve for m and b

We now have the system of equations:

$$\begin{aligned}\sum_{i=1}^n x_i y_i - m \sum_{i=1}^n x_i^2 - b \sum_{i=1}^n x_i &= 0, \\ \sum_{i=1}^n y_i - m \sum_{i=1}^n x_i - nb &= 0.\end{aligned}$$

Solving for b : Rearrange the second equation:

$$b = \frac{\sum y_i - m \sum x_i}{n}.$$

Solving for m : Substituting b into the first equation:

$$\sum x_i y_i - m \sum x_i^2 - \frac{\sum x_i (\sum y_i - m \sum x_i)}{n} = 0.$$

Multiplying by n to clear the fraction:

$$n \sum x_i y_i - mn \sum x_i^2 - \sum x_i \sum y_i + m \sum x_i \sum x_i = 0.$$

Rearrange:

$$m \left(n \sum x_i^2 - (\sum x_i)^2 \right) = n \sum x_i y_i - \sum x_i \sum y_i.$$

Solving for m :

$$m = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2}.$$

Since we already know m in terms of x_i, y_i , we can put the value of m in the equation for b , if we require so. But for simplicity, we let this be:

$$b = \frac{\sum y_i - m \sum x_i}{n}.$$

Final Formulas

Therefore, we have:

$$m = \frac{n \sum x_i y_i - \sum x_i \sum y_i}{n \sum x_i^2 - (\sum x_i)^2},$$
$$b = \frac{\sum y_i - m \sum x_i}{n}.$$

Does the Regression Line Pass Through the Mean of the Data?

To show that the regression line passes through the mean point $(\bar{x}, \bar{y})$, given by:

$$\bar{x} = \frac{\sum x_i}{n}, \quad \bar{y} = \frac{\sum y_i}{n}.$$

we substitute $x = \bar{x}$ into the regression equation:

$$y = mx + b.$$

Using our formula for b :

$$y = m\bar{x} + \left(\frac{\sum y_i - m \sum x_i}{n} \right).$$

Since $\bar{x} = \frac{\sum x_i}{n}$ and $\bar{y} = \frac{\sum y_i}{n}$, rewriting:

$$y = m \cdot \frac{\sum x_i}{n} + \frac{\sum y_i - m \sum x_i}{n} = \frac{\sum y_i}{n}.$$

Since the $m \sum x_i$ terms cancel out, the equation is satisfied. Therefore, the regression line indeed passes through $(\bar{x}, \bar{y})$.